CHAPTER 8

Blockchain technology in biomanufacturing: Current perspective and future challenges

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Abbreviations

ACN	AgreCoin
ACTIV	Accelerating COVID-19 Therapeutic Interventions and Vaccines
AI	Artificial intelligence
CDMO	Contract development and manufacturing company
CFR	Code of Federal Regulation
cGMP	Current good manufacturing practice
CI	Carbon intensity
CMO	Contract manufacturing company
DLT	Distributed ledger technology
DON	Deoxynivalenol
DSCSA	Drug Supply Chain Security Act
EPA	Environmental Protection Agency
FDA	Food and Drug Administration
FHIR	Fast Health Interoperability Records
FMD	Falsified Medicines Directive
GPS	Global positioning system
IBM	International Business Machine
IoMT	Internet of medical things
IoT	Internet of things
ISO	International Organization for Standardization
KPMG	Klynveld Peat Marwick Goerdeler
ML	Machine learning
P2P	Peer to Peer
POC	Proof of concept
RFID	Radio frequency identification
SAP	Systems, applications, and products in data processing
SOP	Standard operating procedure
USDA	United States Department of Agriculture

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1. Introduction

An inevitable transformation has been envisaged from industry 1.0 to industry 4.0 with inculcation of advanced technologic practices commonly known as digitalized manufacturing (Narayanan et al., 2020). The adoption of industry 4.0 has been visualized in various manufacturing sectors, and biotechnology is no longer an exception (Rinker et al., 2021). The increasing realization of the potential of biobased entities for coping with the challenges related to sustainable development, energy requirements, food security, climate change, and productive agriculture has encouraged the scientific community to hasten the pace of biomanufacturing. The inherent advantages of biomanufacturing such as reduced toxic by-products, less fossil fuel consumption, eco-friendliness, and no greenhouse gas emissions necessitate the intervention by the latest technologic developments (Zobel-Roos et al., 2020; Gargalo et al., 2020). In the past decade, biomanufacturing has become equipped via the latest internet of things (IoT) and internet of medical things (IoMT) that have been successfully implemented by enhancing the research diaspora managing and sharing research data for their wider utilization (Kelly et al., 2020; Koutras et al., 2020). In fact, various chemical companies have adopted biomanufacturing for reducing the burden of pollution into the environment as their preferred method of production. The trend is continuously growing for the massive production of biobased products per day. Therefore, there is a dire need to strengthen the arena of biomanufacturing from low cost to high production speed with flexibility in the production capacities as well as to transform consumer's expectations and enhance business capabilities. However, the complexity and high specificity of the biologic system has caught biomanufacturing into a Sisyphean task, and the business and notions of commerce and trade are still a trap of Victorian grandeur. The execution of all business practices is still relying on the intervention of intermediaries, so the development of technology that embraces the glory of security, transparency, immutability, and auditability will definitely transform the height of biomanufacturing. We are living in the era of technology overhaul where digitalization is intervening in private and public domains with lots of information and innovation. Thus, among various technologies, blockchain technology is being heralded as a major breakthrough in the life science and biotechnology sectors, which directly affects the pace of biomanufacturing (Justinia, 2019; Makridakis and Christodoulou, 2019). This technology mainly comprises a peer-to-peer

(P2P) integrated network platform involving mathematical expressions, advanced statistical algorithms, and cryptography for alleviating the discrepancies of conventional distributed database by replacing consensus algorithms. Blockchain has been hailed as a revolutionary digital architecture constituting a shared and immutable ledger having decentralized, open-source, anonymous, and transparent data for public or private organizations (Kuo et al., 2017). Biomanufacturing involves critical steps of involving biologic cells or enzymes to convert various substrates into commercially important products.

This requires diverse avenues for process development for monitoring and control in which blockchain fits for secure and traceable data, for giving the platform of precision and accuracy, as well as providing predictive control for successful execution. The data produced during upstreaming can be well utilized for framing various downstream processes, providing provenance that further paves the way for various collaborative research and development. From inventory to supply of bioproducts, blockchain technology can be well applied for creating the transparent, auditable, and secure data that leads to development of strong relationships with customers. The implementation of blockchain technology has heightened the framework of various business policies to expand the horizon of biomanufacturing that can be directly correlated with bioeconomy (Weking et al., 2020). Fig. 8.1 summarizes the applications of blockchain technology in various sectors contributing the economic growth. The technology also provides a digital backbone for inculcating other advanced technologies such as artificial intelligence, cloud computing, eHealth, and mHealth applications (Lake et al., 2014; Shi et al., 2020; Ding et al., 2019). This digitalized ledger was introduced first through Bitcoin and various other blockchain architectures consisting of consensus algorithms that were developed depending on the specific applications, which have been well cited in the literature. For diversified applications based on access and permissions, three types of blockchain—public, private, and consortium—have been reported. Though the technology has marvelous credentials, it is still evolving and needs to access its full potential in this arena. Thus, the chapter discusses the current applications of emerging blockchain technology in biomanufacturing, current challenges, and future perspectives. The information given in this chapter should not be considered all inclusive, but rather it serves as a foundation in the attempt to give certain glimpses of the future of biomanufacturing.

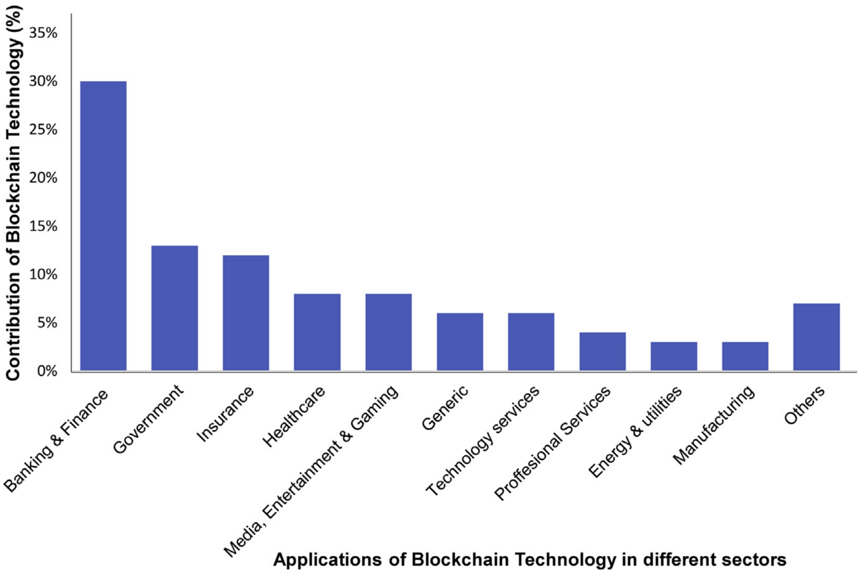

Figure 8.1 Applications of blockchain technology in various sectors and its contribution toward economic growth.

2. Conception and functionality of blockchain

The term blockchain is not clearly defined, but the technology was conceptualized in 2008 under the pseudonym of Satoshi Nakamoto. The technology is designed to facilitate online transfer of money from one party to another without involving the need of any financial organizations or intermediaries. It was introduced as cryptocurrency in the form of Bitcoin (Marella et al., 2020). The timeline diagram in Fig. 8.2 shows the historical development of blockchain technology. The purpose of this technology is to ensure the authenticity and security of data and money transactions, with complete information given to all participants (Ikeda, 2018). It is a kind of integrated P2P networking platform in which each member is designated as a node that can be connected with single or multiple computer systems. After receiving a certain amount of information either in form of data or transaction record or clinical data, patient information, or manufacturing batch number, they become validated and combined to form an immutable "block," and every new bit of information is added in a new block to the existing chain to be continued as a blockchain (Hoy, 2017). Recently, the International Organization for Standardization (ISO) described blockchain technology as a kind of digital platform for storing and verifying the various

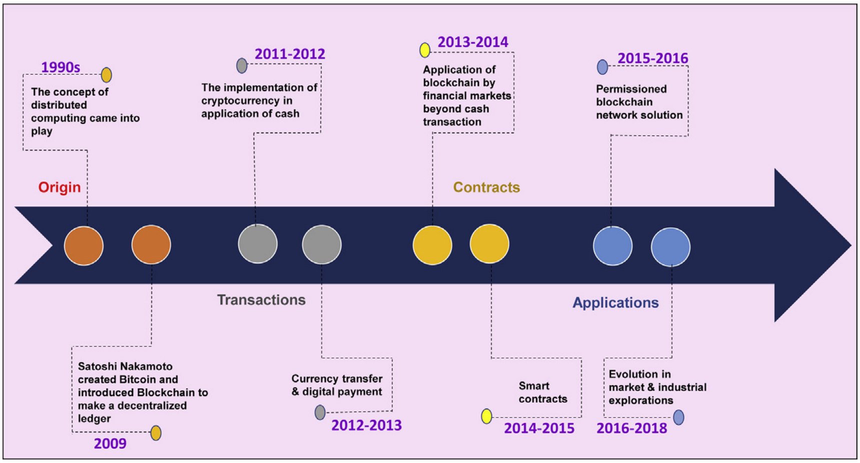

Figure 8.2 Historical development of blockchain technology.

classes of data through a shared, immutable ledger (Tasca and Tessone, 2019). The distributed ledger technology (DLT) is the part of blockchain that opens a door for multiple business operations in various verticals (Kuo et al., 2017). The flexible option for sharing a network among private and public organizations, or the combination of both, enhances its applicability in the wider context. The market size of this technology is expected to grow from USD 3.0 billion in 2020 to USD 39.7 billion by 2025, and the estimated compound annual growth rate is expected to reach 67.3% during 2020–25. The most popular framework of blockchain was launched by Linux foundation in 2016 by the name of Hyperledger. Later on, various other platforms such as Ethereum, R3, BigDB, Ripple, and EOS have been published and applied on various applications per user specifications (Ullah et al., 2020).

A stupendous stride has been seen in the domain of biomanufacturing that is considered the most heterogenous and complex system among the commercialized sectors. However, with the advent of automation, smart equipment, advancements in unit operations of upstream and downstream processes, process engineering tools, and digitalization of batch-to-batch documentation, an enormous amount of data has been generated (Deepa et al., 2020; Yuheng et al., 2020). As biomanufacturing requires stringent regulatory compliances and standardization before market launch, data should be quite secure, authenticated, and reproducible. Therefore, per the specification of US Food and Drug Administration (FDA) the output of the generated data must be attributable, legible, contemporaneous, original, and accurate (Radziwon et al., 2014). The postpandemic era brings the demand for coordinated manufacturing strategies for combating various global issues. Thus, blockchain can be clearly envisioned as a blockbuster technology to achieving such an ambitious project that could never have been executed before its arrival.

3. Why blockchain is needed in biomanufacturing

The arena of biomanufacturing is vast and involves diversified products from low to high value based on market volume. The purity level is a major factor in having bioproducts of therapeutic value. Thus, both manufacturing and purification process comprises critical steps for achieving the desired success (Ahmad et al., 2018; Zhang et al., 2017). The manufacturing involves the fermentation process, culturing of mammalian and higher plant cells, as well as recovery and purification steps that require critical chromatography processes such as gas chromatography, high performance liquid chromatography, gel-exclusion chromatography, affinity chromatography, centrifugation, ultrafiltration, and other advanced purification processes (Aijaz et al., 2018). The diversified layout of bioreactor designs also plays a pivotal role for execution of successful biomanufacturing. Thus, monitoring critical process parameters during the manufacturing is an important step for mitigating the risks of the process. One of the prime challenges to retain the stability and functionality of biologic products depends on the structure of biomolecules and posttranslational modifications in the host. These posttranslational modifications enable the bioproducts to cope with harsh environmental conditions and enabled them in remain in an active state. Therefore, the various aspects mentioned in the arena of biomanufacturing need an automated, transparent, immutable, and digitalized infrastructure to observe and recorded the various process inflows and outflows as well as product distribution in the consumer forum. Thus, by evaluating these requirements, blockchain technology is very fit for this purpose as applied to different facets of biomanufacturing (Zobel-Roos et al., 2020). Though biopharmaceuticals mainly known for their production of biologics are skeptical to imbibe the full potential of this technology, currently, various big names like Bayer, Sun pharma, Pfizer, Sanofi, Johnson & Johnson, and others have adopted this technology in their framework (Xu et al., 2020; Haq and Muselemu, 2018). It is clearly envisioned that though the technology is not the solution for a panacea of problems in biomanufacturing, with the integration of advanced digitalized technologies, it should be a boon for biomanufacturing and provides a more comprehensive and transparent fast-track platform from the manufacturer to consumer perspectives (Gargalo et al., 2020). The key features of blockchain technology used in biomanufacturing are summarized in Table 8.1. This will certainly accelerate the production of biobased commodities and contribute prolifically in the growth of bioeconomy and sustainable development.

4. Key elements of blockchain that are required in biomanufacturing

Blockchain technology offers nonrepudiation of transactions across a distributed P2P network in the absence of intermediary sources. The world of biomanufacturing is also warming up to the innovativeness of blockchain technology. The biomanufacturing industries can share and integrate the data of various capital, operating, and administrative costs for setting up the industry. The immutable information on the purchased cost of equipment and specification of various bioreactors and downstream processing equipment can also provide trusted information for assessing the quality attributes that have been used for manufacturing purpose. This technology offers various features that can be a transformative factor for broadening the horizon of biomanufacturing. The mind map represented in Fig. 8.3 depicts various facets of blockchain in biomanufacturing. The key elements of blockchain that play a crucial role in developing smart manufacturing are described subsequently.

4.1 Data governance and management of records

The domain of biomanufacturing comprises various regulatory issues in terms of ethical and social acceptability (Quaye et al., 2009). The specificity of biobased products and utilization for therapeutic and food purposes posits a stringent level of regulation. Thus, significant burden has been seen among consumers and manufacturers for building trust and developing a long-term relationship by keeping records and making legal documents. Therefore, unimaginable possibility can be conferred by blockchain technology to maintain the data and associated records in a secured and encrypted manner in a digitalized framework. This will facilitate the smart contract between companies and consumers that builds trust and offer more opportunities for the growth of biomanufacturing business (Paliwal et al., 2020).

Table 8.1 Key features of blockchain technology that can be implied in the biomanufacturing arena.

S.No.	Characteristics of blockchain	Functionality description	Its benefits in biomanufacturing
1.	Autonomy	This means that each node of blockchain can access, transfer, store, and update the date securely.	• Data can also be shared with public sectors securely to reap more benefits.
2.	Decentralized	This means that the data can be accessed, monitored, stored and updated on several computers.	• A manufacturer can access the data and monitor the work and if there is any diversion from the given protocol. • It will be easy to trace and track the supply chain.
3.	Transparent	This means that the data is transparent to its user; it helps in preventing data from being altered or stolen.	• Consumer and manufacturers will develop a transparent relationship with more trust. • A quality check can also be possible in every step.
4.	Immutable	This means that the blockchain ledger remains unchanged, and the data stored once is reserved always in the form of a specified hash value.	• Products cannot be counterfeited in the journey. • Any changes in manufacturing conditions can be kept on record.
5.	Open-source	This means that blockchain has open-source access to all of its connected networks.	• Consumers can have access to the manufacturing processes to be assured of the quality of the product.
6.	Anonymity	This means that as the data is being transferred from one node to another node, the identity remain anonymous.	• Stringent conditions required by several bioproducts could be keep in check on a regular basis by various organizations.

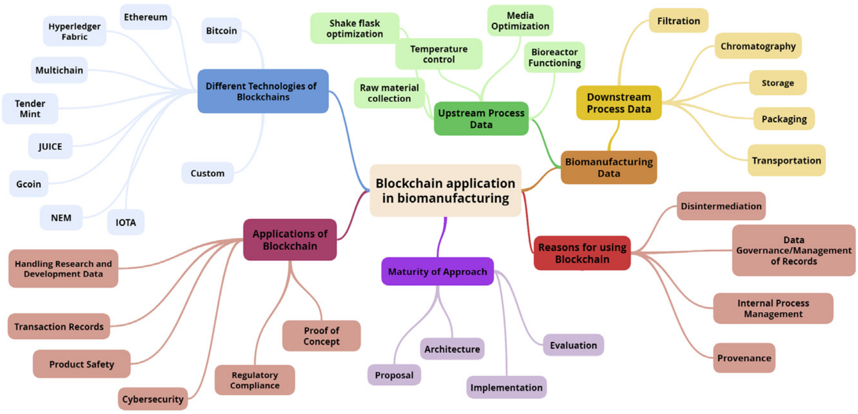

Figure 8.3 A mind map depicting the various avenues of blockchain technology in biomanufacturing.

4.2 Disintermediation

The major strength of blockchain lies with the nonrequirement of intermediaries. This will save cost and time for companies and obviate fake and fraudulent information that can posits risks for business management. The transmission of secured information to all the parties whether in terms of private or public will develop the trust among networking partners and facilitate a collaborative environment (Tan et al., 2021; Giannini and Terenzi, 2020). The trials on assessing the therapeutic efficiency of monoclonal antibodies imparts a successful example in which investigators, regulators, manufacturer and trial sites can be connected with one digitalized platform and all data can be shared in encrypted manner.

4.3 Internal process management

The manufacturing sector has its own internal system to manage the raw materials, factory operations, product processing, waste management, packaging, and labeling of the products. Each domain requires a separate record of transaction among the internal members of the organization (Mendling et al., 2018). The blockchain provides a holistic approach to reconcile all transactions through a single shared ledger. This will enhance the transparency in every domain of the manufacturing followed by the development of trust among employees for making a better environment.

4.4 Provenance

The attribute of tracing the origin and authenticity of the various objects in the manufacturing sector is known as provenance. The manufacturing sector has to face various counterfeit trading objects from supply of raw material to out-of-order equipment. The process modification also affects the batch-to-bath variability. Therefore, for keeping the records in immutable form, blockchain provides a magnificent platform for tracing all the inventory from process manufacturing and product packaging to transportation. Inconsistencies can be traced due to the interconnection of blocks with their predecessors in encrypted code.

4.5 Handling research and development data

The biggest challenge among biomanufacturing is the reproducibility of the process, as the biology has inherent complexity in terms of cellular framework as well as in products profiles. Therefore, rigorous research must be needed periodically to upgrade the system and assess its reproducibly. These tasks require considerable money, which is a major roadblock in small-scale and startup biomanufacturing sectors. Therefore, any malicious access to research and development data will impact the reputation in terms of funding and consumer relationship of an organization (Wang et al., 2019). Thus, blockchain technology provides a wonderful platform for enabling the access of sensitive research data to the real users with the consensus of trusted parties and investigators. Thus, to leverage the growth of biomanufacturing, an ecosystem with generation and sharing of data must be required nowadays to fulfill the rising demands of biobased commodities (Naz et al., 2019; Ozercan et al., 2018).

5. Various avenues of utility of blockchain technology in biomanufacturing

5.1 Infrastructure development

The infrastructure of biomanufacturing has to carefully comply with cGMP practices and other stringent containment regulations. The clean room facilities and stringent regulation of temperature in biologic reactions require careful monitoring of data in secured and nonauditable form to prevent any discrepancies in the biologic process.

The capital cost of running the industry and operation cost for different utilities required to manufacture the products constitute an important segment of infrastructure. Thus, the documentation of cost and maintenance, audits, and regulatory framework requires extensive data management for running a smooth and cost-effective process. All these requirements can be leveraged by blockchain technology where transparent and secure data can be shared to desired parties in immutable form, and modification can only be done by consent from all involved parties (Rui et al., 2019). The technology is flexible enough to work on both privatized and public forums, or combination of both, which adds more dynamism. The past decade has seen tremendous efforts to develop various platforms from recording the data of raw materials to manufacturing and transportation to end users (Darvazeh et al., 2020). In 2017, the joint efforts of 15 pharma and biopharmaceutical companies enabled the development of SAP Pharma blockchain POC (proof of concept) app with collaboration of software giant SAP for tracking the supply chain management of various drugs and biologics, and it currently adds more technology features to upgrade with advanced facilities. The rising popularity of this technology in the biotechnology industry facilitated the development SAP Information Collaboration Hub for Life Sciences in 2019. This blockchain-based platform provides the connection of a customer forum directly complied with the US Drug Supply Chain Security Act (DSCSA). The discrepancies in drug manufacturing can also be tracked by inclusion of this technology.

Recently, Novartis created a blockchain technology-based solution for tracking counterfeit medicines. Along similar lines, Merck also developed its own technology platform for enhancing the security of its supply chain (Jennings, 2021; Morris, 2018). Moreover, Boehringer Ingelheim (Canada) collaborated with IBM for improvements in data sharing and storage of various clinical trials and patient data. In addition to that, the joint efforts of Amgen, Sanofi, and Pfizer developed a trial platform using blockchain technology to optimize the process for cost-effective drug production. The more powerful platform based on this technology was developed by Exochain for allowing patients to control the interaction and usage of their medical data by the research community, which leads to further the trust development among various parties (Mackey et al., 2019). MediLedger developed, by collaborative effort of 20 companies, has been used as an interoperable system constituting multiple parties, including manufacturers, wholesale distributors, hospitals, and pharmacies for verification and transfer of various biologic and chemical drugs with trust and authenticity (MediLedger, 2017). The global supply chain platform was developed by the joint effort of IBM with KPMG, Merck, and Walmart for efficient supply of biologic drugs (Miller, 2019). Though the current examples have been more directed toward supply chain management, which is an essential part of infrastructure development, various blockchain platforms are under development that can be exclusively used for monitoring the manufacture of biobased products. The increasing realization of potential of gene and immunotherapies leads to the development of personalized medicines. Thus, the rising trend of production of very high-value, low-cost, biotherapeutic products utilizes a single-use bioreactor strategy (Jacquemart et al., 2016; Bandyopadhyay et al., 2017). Monitoring various biologic parameters and recording the data is a secure manner facilitates data sharing and tracking opportunity for customers and other verifiers and regulators, which enhances the visibility and trust of the product and industry. The viral contamination of high-value products manufactured from mammalian cell culture techniques requires extensive column chromatographic techniques, so by integrating the process with blockchain platform, the data can be used for continuous monitoring and shared for its verification and authenticity. Apart from that, this technology has provided a magnificent platform for Contract Research Development Organization (CMO/CDMO) sharing the research and development data in a secured manner. The infrastructure development of 3D printing of personalized drugs requires an advanced manufacturing system that can be integrated with blockchain and machine learning platforms to analyze the secured data for assessing the correct dosage and formulation of drug, depending on the profile of patients, as well as its production.

5.2 Advancing machines as a service

Biomanufacturing involves the complex process of upstream and downstream processing for harnessing the products from living cells. This process requires a different class of machine from production to purification. Currently, the concept of innovative pay-per-use model for machinery has been more prevalent and is known as "machines as a service" (Lillrank and Särkkä, 2011). The model works on the principle of providing the service of equipment or machine use on the basis of the generated output on a charged basis that curtails large investments and gains accesses to upgraded, latest technology. The model also provides flexibility to enhance the production capacity of manufacturers and makes the arduous scale-up process cost-effective. The model can only be used when this process is integrated with a tamper-free, secured, and shared data platform, and blockchain provides feasible option for the implementation (Machine-as-a-Service, 2018).

5.3 Enabling machine-controlled maintenance

An advanced biomanufacturing facility requires automated service agreements and shorter maintenance times. The technologic sophistication and the health of each piece of equipment can be monitored and maintained by appending the installation documentation and service agreements in the device through a blockchain platform for the creation of a digital twin (Zobel-Roos et al., 2020). By integrating with a blockchain platform, an automated execution and payment of scheduled maintenance can be done in a secured way. The equipment that needs maintenance itself can trigger a request for servicing and create a smart contract for the work execution or replacement part. After accomplishment of work, automatic processing of payments happens, and history is appended in nonauditable documents connected with a blockchain platform. This enhances the reliability and reproducibility of the utility of the correct equipment for the manufacturing. Though this concept is in its development phase, it enables the reduction of cost and maximizes profits for manufacturers without compromising credibility or trust (Justinia, 2019; Shi et al., 2020).

5.4 IP and tech transfer

There is considerable gap in technology transfer from academia to industry that confers a strong roadblock for wider dissemination of research among the scientific community. The timely delivery of research is a tedious process due to difficulty in identification of potential opportunities and management of data among collaborating scientists from different institutions and industries. Therefore, the research data generated during research and development of biologic products should be carefully stored and secured to develop a chain, and updated data will be added time to time for attaining transparency. Thus, blockchain-based platforms are well suited for the management of technology information and tracing the research and development work through a decentralized and convoluted platform (Qureshi and Megías Jiménez, 2020). This platform is also well suited for investors for identification of required owners and recording deals, regulations, and payments for future use. This technology paves the way for achieving milestones in protecting intellectual property and license agreements between interested parties in a digitalized form.

5.5 Inventory management

Biologic products can be made from a wide range of raw materials that can be available locally or imported. Natural and waste bioprocessing requires critical inventory management due to fluctuation in the availability of raw materials that further leads to batch-to-batch variation. The stock of raw materials and their maintenance is coming under the umbrella of good practices. The regulatory framework and financial aspects need to be carefully stored and monitored for their stability and storage. Thus, for accomplishing this task, blockchain technology is a sustainable option for maintaining consignment stocks globally, enabling the tracking of the location and custody of materials in a secured and immutable way (Astarita et al., 2019).

5.6 Simplifying and safeguarding quality checks

Currently, biomanufacturing has been recognized as the factory of the future due to highly specificity of bioproducts and an ecofriendly approach toward the environment. The inherent property of blockchain platform for creating the immutable documents for the production and supply chain data of bioproducts can be used for quality checks and verification of these products at different stages of manufacturing. This application can only be feasible when automated quality checks have been done with accurate measurements and their documentation through blockchain. The information can be further utilized by a controlling authority for the timely update of data, reducing the need of auditing by quality control personnel and other manual interventions (How blockchain, 2019).

5.7 Serialization of bioproducts

The arrival of gene therapies has led to a devastating impact on the biomanufacturing sector. In 2020, four gene therapies were approved by the US FDA, and 900 therapies are estimated to be in the pipeline. It is clearly envisioned that there is need of serialization to enhance the business opportunities at a global scale through standardized supply chain management. The process of serialization involves giving a unique identity number of each biologic or drug that is used to decode the batch number, location, date, and its logistic route and manufacturer that can be shared with supply chain trackers as well as with consumers. The objective of this process is to trace the products from their origin.

The security and transparency of blockchain technology is to comply with the industry's standard serialization regulations that can be applicable throughout the globe. Till date, two major governing bodies, US DSCSA and EU Falsified Medicines Directive 2011/62/EU (FMD), have been acknowledged as certified serialization organizations, and blockchain technology readily complies with these organizations to manage the supply chain in an efficient manner (Naughton, 2017, 2018). Currently, serialization process is in great demand to boost the dimensions of business, and recently the company Servier, which has spread in 148 countries and is a major player in serialization and global supply chain, developed a blockchain-based platform that is connecting the biomanufacturing hubs of different countries. Thus, blockchain technology makes the biomanufacturing sector more recognizable through the interconnected industries, opening the door for more collaborative developments in this field.

5.8 Bioproducts expiration and bioproducts recall

The panorama of biobased products is vast and various generalized to specific products are manufactured through either whole cell or immobilized cell and enzymes through fermentation as well as biotransformation. The stringent operations have been more directed toward those bioproducts that can be used for therapeutic purpose. Various cellular products like interferons, interleukins, growth hormones, insulin, monoclonal antibodies, embryonic stem cells, and vaccines have been used for therapeutic interventions. These products require sophisticated control of temperature and humidity and have a shelf life. Though utmost care has been taken for biomanufacturing of these products, sometimes inevitable conditions lead to the deterioration of product, which may lead to life-threatening conditions for the patient. Thus, with the help of blockchain technology the exact identification and tracking of expired bioproducts can be done while applying the principles of reverse logistics for complete disposal of these products based on standard operating procedures (SOPs) (Sahoo and Halder, 2020). In addition, the other side of this coin is that future recalling of bioproducts due to any discrepancy in manufacturing, packaging, or conferring conspicuous side effects can be easily done by implementing this technology. This technology provides the correct location and chain of custody information by tracking product, batch number, and manufacturer information in a legitimate way (Subramanian et al., 2020).

6. Applications of blockchain technology in biomanufacturing (case studies)

6.1 Blockchain in alcohol biomanufacturing: a case study

Alcohol production is one of the most historical processes in the world of biomanufacturing. The consumption of alcohol is as old as humanity itself.

There is a continuous rise in the consumption of alcohol, and in the future, there is no hope for giving up this habit in spite of imposing many prohibitions. The biomanufacturing capacity of alcohols and alcohol-based beverages is growing at an undeniable high pace and is expected to rise to the $2 trillion mark by the year 2025 (Industrial Alcohol, 2020). The contribution of alcohol production and sales contributes high revenue generation for all nations. The pricing of all alcoholic beverages has been determined by the cost of production and the duties levied on those costs. Therefore, developing a smart manufacturing of alcohol is urgently required for meeting the demand of the rising population. Blockchain technology is a boon for marketing and supply chain management of alcoholic beverages. as 30%–60% of alcohol is fake nowadays. To identify counterfeit product and curb the need for traditional advertising formats for promotion of a variety of alcoholic beverages, this technology plays significant role in the global context. Recently, technology was utilized in the smart and shared information known as AgreCoin (ACN) on feed stock (agave biomass) availability for tequila production. ACN comprehensively known as an Agricultural Agreement Coin, based on an Ethereum ERC20 blockchain platform, is a P2P solution for executing the trade and financial management of agricultural alcoholic products and services (Blockchain Benefits, 2019). In the United States, $2.6 billion of agave biomass is produced annually for the commercial manufacturing of tequila, but the slow growth of the plants (approximately 6 years) leads to a supply crisis of agave that confers a dilemmatic situation among farmers when the product is required at high demand (Enescu et al., 2020; Davis et al., 2010). Therefore, this platform has been used to resolve the agave crisis and save a billion-dollar industry from huge losses. The same concept can be well correlated with beer manufacturing from barley and wine from grapes and other feedstocks. Thus, this platform channelizes the transparent flow of information from producers to consumers and various stakeholders. Apart from that, this concept can be proposed for downstream development of alcohol. The distillation process leads to various products from alcohol, so a digitalized platform can be used for tracking this process for getting authentic products from the manufacturer. The story of the other side of this platform is also worth mentioning here: ACN also facilitates a social impact plan that contributes the utilization of funds for providing free education for underprivileged students of the Mexican community and mining operation in Guadalajara, Mexico (Oñiguez et al., 2014).

6.2 Blockchain in the development of COVID-19 vaccine: a case study

The whole world is scourged by the catastrophe of the ongoing COVID-19 pandemic that confers devastating impact on economic, social, and public health at a global level. This situation witnessed a lot of loopholes in current healthcare management systems that lack the transparency, security, and fast tracking of data for efficient management. However, a silver lining growth has been envisaged in the biopharmaceuticals for the development of vaccine in a global context to defeat the outbreak of SARS-CoV-2 (AgreCoin, 2018). The great efforts of vaccine development have been done throughout a global context by a collaborative approach of the scientific community and biotechnology industries, rather than a competitive one. Thus, it would require an unprecedented level of manufacturing and distribution capacity. It is estimated that approximately seven billion vaccine doses will be required to suffice the need of every age group in all continents. In addition to that, it has been also estimated that 20%–30% loss will occur during transport and distribution, so approximately 10 billion doses will be needed, and furthermore, keeping in view of the requirement of two doses of vaccine per individual, 19 billion vaccine doses will be required to stop the further spread of the virus (Calina et al., 2020). Thus, it is doubtless simple to comprehend that there is equitable role of manufacturing as well as supply chain management. Thus, there is strong upsurge in the development of a decentralized, distributed blockchain technology that can provide real-time monitoring of secure and encrypted data without alteration of real information that leads to the development of trust without the involvement of a third party that encumbers the risk as well as fraud and fake information related to vaccine development and distribution. The year 2020 witnessed the prolific development of various blockchain software assisted by artificial intelligence and IoT platforms to overcome the discrepancies in the process of defeating the pandemic. The development of vaccine requires prodigious research on the mechanism of virus replication, attachment, and its binding mechanism. However, the clinical data from the patients and their symptoms provide a broader understanding of the infection outcome. Thus, various research projects at a pilot scale comprising blockchain technology have been implemented for collecting and storing the data on a wider public forum. Recently, a public–private partnership has been reported to develop a framework to accelerate the development of vaccine and drug molecules, streamlined clinical trials, and coordinated regulatory compliances, known as "the Accelerating COVID-19 Therapeutic Interventions and Vaccines (ACTIV)," which is based on blockchain technology. Apart from that, various research projects implemented at a gross scale have been reported, and some of the examples are listed.

6.3 MiPasa platform for preventing false infodemics in the public forum

The management of COVID-19 pandemic needs authentic and verifiable data for creating policies for governments and industrialists. MiPasa is a blockchain technologic platform based on Hyperledger fabric that is connected with authentic sources like WHO, registered health authorities, and organizations for relaying the information of virus spread and their outcome in the public forum (Kim et al., 2021). This authentic and immutable information is further utilized by the government and private authorities to develop various pharmacologic and nonpharmacologic solutions for eradicating the spread of the virus. This platform is also helpful for identifying COVID-19 carriers and infection hotspots. Similarly, Hashlog is a blockchain-based platform that provides data about the COVID-19 outbreak integrated with WHO and ECDC databases.

6.4 VIRI platform for preventing spread of COVID-19

The identification of infection hotspots of the virus through digital contact tracing has proven to be a milestone for limiting the spread of airborne infection of COVID-19. Data privacy as well as authenticity are two key issues that are able to solved by the blockchain-based digitalized VIRI platform. In fact, the wider applicability of this platform for identification of virus carriers across different countries has given the idea of the fast viral emergence and outbreak. This platform fully preserves the user's data and is integrated with artificial intelligence (AI) and machine learning (ML) tools that can be further used for the accurate prediction of COVID-19 pandemic in a global context (Salah et al., 2020).

6.5 WIShelter platform for assurance of privacy of COVID-19 data

WIShelter is smart phone–based application originating from Wisekey's digital identity platform for providing secured data to consumers. The clinical data both from healthy and diseased individuals is stored in a secured and encrypted manner through a blockchain technologic platform. This forum can also be equipped with uploading a COVID-19 testing certificate, which is information can be further used to decide the home quarantine possibilities or hospitalization. This application is also helpful for deciding travel requirements of the public based on their data availability and situation.

6.6 Blockchain in the development of biofuels: a case study

The realization of depletion of fossil fuels in the 1970 and 1980s propelled the scientific community to find out a sustainable alternative to fulfill the rising demand for energy. The development of bioethanol from corn as feedstock led to the groundbreaking step for the development of biofuel in various countries across the globe (VIRI Creates, 2020). Brazil has been in the forefront in this development, but this is not yet highly used because of using land and food crops for the production of biofuel. This approach is called first-generation bioethanol, but later on, biofuel development from waste lignocellulosic biomass laid the foundation for second-generation biofuels (Bušić et al., 2018). The inherent complexity of lignocellulosic biomass is imposing considerable challenges for its commercialization in spite of the technology being ecofriendly and reducing the carbon footprint in the environment. Copious research has been carried for consolidated bioprocessing for the conversion of lignocellulosic biomass by using advanced molecular biology, transcriptomics, proteomics, and system and synthetic biology approaches and has paved the way for establishment of biorefineries for harnessing the maximum possible value-added products (Robak and Balcerek, 2018). Thus, it is a dire need to establish a tracing mechanism and to store the information in secured manner from "feedstock to fuel." Various countries are adopting the technology of biofuel for reducing the burden of pollution and attaining sustainability. The prime challenges in this paradigmatic shift are the stringent regulation, unpredictability of feedstock type and availability, thinner margins, and demand of transparency from the consumers' side. Thus, blockchain technology is indeed a suitable platform for addressing these challenges by developing a "digital passport" for every process of biofuel development (like types of feed stock, availability, storage, process development, sugar content, production information, ignition value, and supply chain) (Pathak et al., 2019; Andoni et al., 2019). The prospective applications of blockchain technology in the biorefinery sector are represented in Fig. 8.4. This information should be stored in separate blocks and added in a chain that can be encrypted with digital code, so information cannot be forged, giving

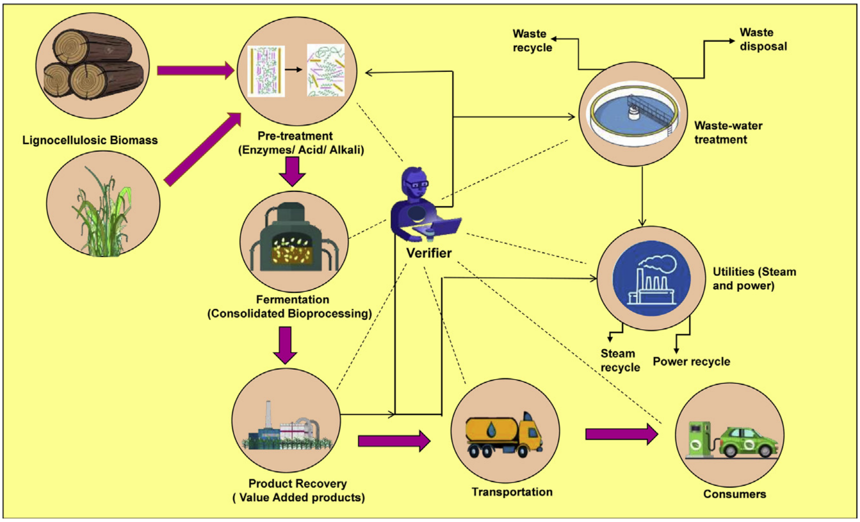

Figure 8.4 Futuristic application of blockchain technology in the biorefinery sector.

transparent and updated information. The most critical parameter for adopting biofuels is carbon intensity (CI). The fuel quality would be good if it achieves lower CI, which is possible only by growing the feed stocks in a reduced amount of nitrogen fertilizers. The farmers should be incentivized for using the lower amount of nitrogen technology for the growth of crops by adopting engineered nitrogen-producing microbes and slow-release nitrogen nanotechnology. These practices reduced the burden of nitrogen fertilizers that further impact the low CI of biofuel. The blockchain platform is also use for tracing the contamination of feedstock with toxins like vomitoxin (DON), which can prevent million-dollar losses in this industry (Wu and Tran, 2018). Recently, Grain Discovery blockchain platform developed in Canada has been used for resolving the problem of DON contamination in the ethanol supply chain. The test data has been added by corn sellers to the crop's "digital passport," which includes the concentration level of DON from testing in labs. If found in critical concertation, their sale will be rejected, further saving manufacturing losses in the industry (Lane, 2019).

7. Blockchain technology as a prospective tool for bioeconomy

The growth of biomanufacturing is directly corelated with bioeconomy. Currently, the bioeconomy is precisely understood by conjuring the images of recombinant products recovered from genetically modified organisms and biorefineries. The development of biofuels like bioethanol, biodiesel, biohydrogen, etc., from waste biomass and diversified value-added products retrieved from waste valorization is the key technology behind the setup of biorefineries (Becker and Whittmann, 2019). The massive utilization of these products in mundane activities can only be accomplished with legislation and intervention of various government policies. Thus, for a sustainable bioeconomy, the marketing of biobased products will have to be done by highlighting their superiority over petroleum-based commodities in terms of carbon footprint. The feed stock that has been utilized for establishment of biorefineries is also independent of direct land use and based on nonedible crops. The stringent regulatory setup of government authorities also imposes a strong check on falsified news and low-quality manufacturing. Thus, to overcome these limitations, blockchain technology can be relied on to maintain a single flow of secured information in digitalized document forms containing all necessary information from emission profiles of biofuels, carbon benefits, water utility, waste disposal regimes, and valorization process and their value-added products and, most importantly, sustainability credentials. This would alleviate the need for authorities to check compliance records and other stakeholders as well as other "black box" sections. Thus, this model holds tremendous potential that can help build biorefineries as factories of the future. Recently, GoodFuels, a leading company in developing sustainable marine fuels, tested the world's first zero-emission marine fuel. The company announced the use of a blockchain model to store carbon footprints and emission savings for capturing the market of cleaner fuels by building trust, transparency, and a financially sustainable platform for zero-carbon shipping.

8. Current progress

The remembrance of an old adage is quite relevant in the current context: "If you have a hammer, everything looks like a nail." This statement highlights the prominent role of technology for addressing every problem, but the hammer needs to be changed based on latest trends and developments. Blockchain should be considered revolutionary technology possessing the credentials of security, transparency, and digitalization (Justinia, 2019; Makridakis and Christodoulou, 2019; Ikeda, 2018). There is plenty of research work that has been translated at a commercialized scale by adopting blockchain technology, but the vertical of biomanufacturing is quite complex and challenging. As the world is moving toward digitalization at a fast pace, many developments have been seen in this sector also. However, the legal, social, and regulatory compliances of exploring and commercializing the biologic world are quite arduous. The technology is not acceptable in the industrial forum unless its credibility has been proven across a national or international context. Thus, for proving the credibility of blockchain technology in the biomanufacturing sector, various technical standards and guidelines should be formulated by government officials, the scientific community, and industrialists that can be considered SOPs for massive utilization. However, the development in this is under progress, and some of the currently published standards are based on the Open Standard for Finance and the Enterprise Ethereum chain alliance (Dewey et al., 2019). For supply chain management, guidelines should be adopted from global supply chain standards (GS1) for attaining uniformity in identity, storage, and sharing of supply chain data. The sharing of clinical data should be channelized through the framework of HL7 (Health Level Seven International) standards for electronic health management systems. The more prolific blockchain-based platform was developed for recording electronic health data in a secured manner that is based on FHIRChain (Fast Health Interoperability Records + blockchain) (Zhang et al., 2018). More emphasis should be laid on the Title 21 Code of Federal Regulations that should be adopted in the biomanufacturing sector. The implementation of blockchain technology for high-value, low-volume products should originate from the guidelines of the US Environmental Protection Agency, the US FDA, and the US Department of Agriculture (Kamilaris et al., 2019). Currently, big pharmaceutical giants Biogen, Pfizer, Merck, AstraZeneca, Deloitte, and GlaxoSmithKline have developed their supply chain framework based on blockchain. Apart from that, the amalgamation of blockchain technology with location-based technologies such as GPS, RFID tags, and Sigfox has been envisaged for the development of automated platforms where manufactured products themselves tell us where they are, what condition they are in, who has received them, and whether they have been paid for. However, blockchain is amenable to be upgraded with technologic frameworks like 5G, AI, and ML for more comprehensive applications at a successful rate.

9. Proposed model of biomanufacturing through "FabRec" platform

Blockchain technology is still too much in its infancy to be comprehensively applied in biomanufacturing through connected machines and their accessories to develop decentralized manufacturing ecosystems. The technology should be equipped with proper authentication for recording the identity of machines and the process flow in a distributed ledger that requires verification of various nodes and their roles and responsibilities. Apart from that, the technology will also assist the design of layouts for interacting networks with their different levels of participants in manufacturing marketplaces. A prototype of a smart manufacturing ecosystem that was developed based on blockchain technology is the FabRec platform (Angrish et al., 2018). It is a framework in which a decentralized network of customers and manufacturers can interact through a laboratory-based setup of nodes and system-on-chip board for visualizing the integrity of data and process workflow for the development of trust. Here, FabRec is proposed for a decentralized framework for the development of a biomanufacturing ecosystem. The overview of the proposed model based on FabRec that can be applied in biomanufacturing sector has been represented in Fig. 8.5. The model depicts the integration of nodes that represent the participants, which can be human or machine, that can be interact with each other through blocks. Each block has all the relevant information of manufacturers that can be located in remote areas or developed areas, and the information can be shared through encrypted codes to the users or customers. The dashed lines represent the information inflow among trusted

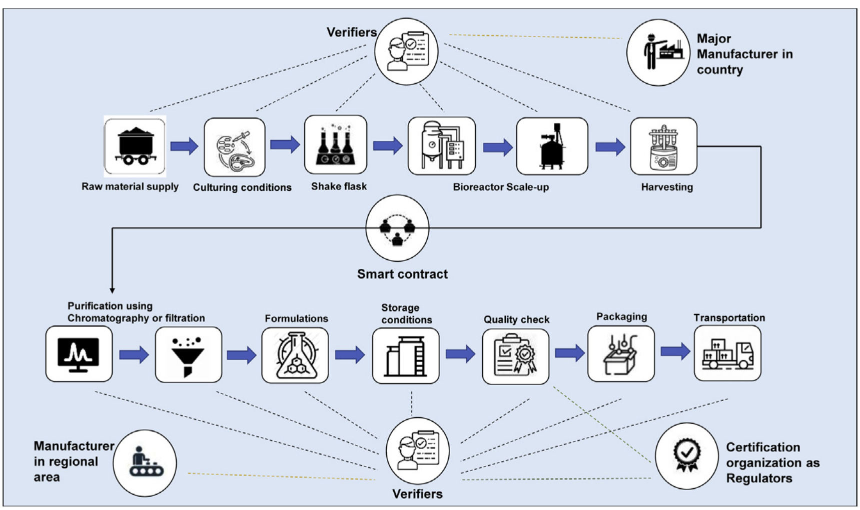

Figure 8.5 "FabRech" platform–based proposed model of blockchain technology in biomanufacturing.

members. Every participant has been allocated with a unique identity, and a diverse arena of people from both private and public sector can join this network. The quality control community such as ISO can join and validate claims conferred by a manufacturer. This platform also provides flexibility of joining the regulators and compliance agents that can help in further verification of biobased products. Thus, this model provides a layout for the development of a smart and digitalized integrated platform for consumer and manufacturers in the biomanufacturing sector. This will certainly play a game-changing role in this sector. However, the true potential needs to be evaluated at a wider scale for its commercialization.

10. Current challenges

Blockchain technology seems promising in the biomanufacturing sector, but it is not a panacea for all the problems, having its own challenges from conception to implementation. As the canopy of blockchain is composed of secured and shared information, it requires the networking of millions of computers at a global level. This networking requires consumption of a huge amount of electricity (terawatt hours) that encumbers the energy consumption of many countries and contributes to a larger carbon footprint. The benefits and challenges of blockchain technology are represented in Fig. 8.6. Another challenge is that although the information will

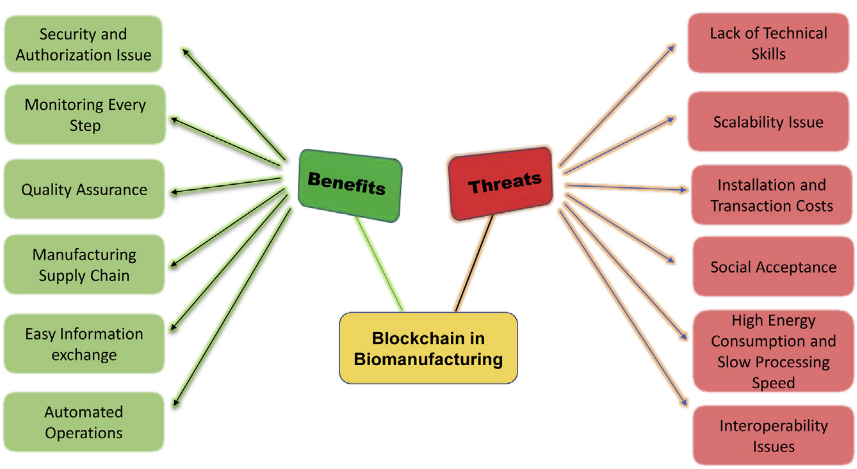

Figure 8.6 Benefits and threats of blockchain technology in the biomanufacturing sector.

be not tampered with after adding to the blockchain platform, the data originator can enter falsified information that is used by other participating members.

Thus, losing the credibility of organization or manufacturer also impacts the wider adoption of this emerging technology. The technology is still lagging behind global usage in the manufacturing sector due to the less attention being paid toward formulating the associated guidelines and regulations. Currently, with the ongoing advancements in bioprocessing 4.0, the research and manufacturing of biologic products create enormous data, but industries are always skeptical to share their data due to security. Therefore, certain commercialized blockchain-based platforms should be developed that can be exclusively utilized in this sector. The technology is new and various applications have been created at a high rate. Proving the credibility of these applications is another daunting task in the biomanufacturing sector. Thus, one can envision that the technology will prove to transformative in the biomanufacturing area, but prodigious research and development should be done prior to implementation of handling sensitive biologic data for global use.

11. Future prospects

The emerging application of blockchain will definitely intervene all facets of life and lead toward the transformation for betterment in the future, the potential in the biomanufacturing sector of this technology still needs to be assessed. Currently, in major enterprises, blockchain projects are in experimentation phase, and the technology faces a "trough of disillusionment." The arrival of the World Wide Web in 1990 faced a similar challenge for global acceptance. So, we can foresee that there will be various interoperable blockchain platforms with easy access, greater scale of operation, and a customer-friendly approach.

The current advantages such as immutability, transparency, and P2P networking in both private and public mode cannot be ignored, but it is the cost of implementation and global acceptance that need to be worked out in a futuristic scenario. It is envisioned that the future of blockchain technology in biomanufacturing will move into a bidirectional mode. The future prospects of blockchain technology in the biomanufacturing arena are represented in Fig. 8.7. One direction of development and improvements lies in developing IoT-based super secured networks for sharing the research and development data in the biomanufacturing sector.

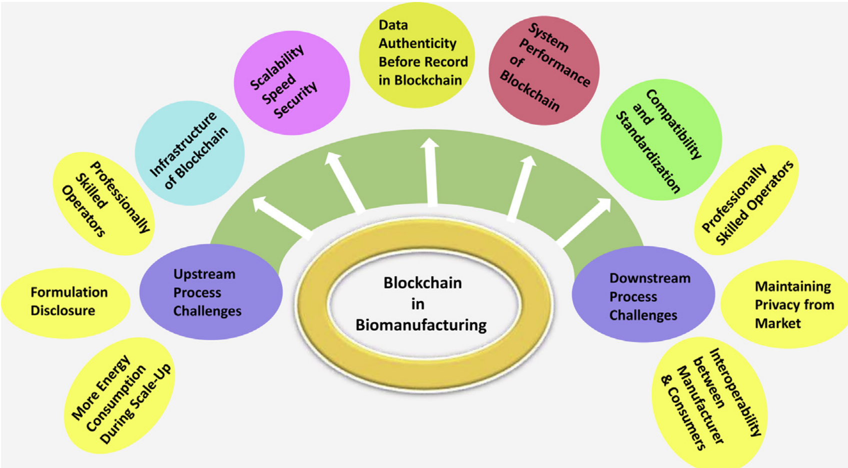

Figure 8.7 Future prospects of blockchain technology in the biomanufacturing sector.

Another direction includes the combination of AI with blockchain technology to upgrade the design of bioprocess, its sustainability, and economic analysis. Furthermore, the cyber security for protecting confidential data of research and development and sensitive data about various pathogenic microorganisms and their process should be also considerable in the future to promote the trust and credibility of this technology in the biomanufacturing sector.

12. Conclusion

Blockchain technology is a revolutionary technology that make the world a smaller place. The applications have started with the financial sector, but its roots are spreading to all business and service sectors. The burst in digitalization and automation accentuates the prolific growth in the development of blockchain technology. The technology has been well embraced in the pharmaceutical sector to maintain effective supply chain management, eradicating the network of counterfeit drugs, but it is nascent in the biomanufacturing sector. As bioprocessing 4.0 is a major technology breakthrough for advanced manufacturing practices, blockchain technology will play an outstanding role in the transformation of this sector. The technology has its own limitations for implementation, but considering its astonishing credentials, it will be an integral part of the bioeconomy in the future. Current advancements in medical devices and diagnostics, tissue engineering and regeneration, biotherapeutics and stem cell therapy, and operative management planning will continue to develop various blockchain platforms that can facilitate biomanufacturing in more secured and transparent way.

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